

Practice Exam AE4135 Rotor and Wake Aerodynamics

Question 1:

Explain the origin of the root vortex as it relates to tip speed ratio, force distribution on the actuator disk and the distribution of circulation on the actuator disk and wake.

Question 2:

Consider a Horizontal Axis Wind Turbine in axial flow, with an infinite number of blades, rotating at a finite tip-speed ratio.

Determine the speed or rotation of the wake at the location of the tip vortices, at infinity downwind. Describe your rationale, and express the velocity in terms of the variables that define the mode of operation of the turbine. If you can think of it, consider different formulations. Describe the different assumptions you use.

Determine the relation between the axial velocity of the wake and the rotational velocity of the wake.

Question 3:

Compare and evaluate the difference between a Blade Element Method and a Lifting Line vortex model for a Horizontal Axis Wind Turbine and for a propeller. Discuss fundamentals of the formulation and limits of applicability.

Describe the implementation of both methods (summary of main aspects).

Question 4:

Name two aerodynamic devices that are being used to tune the blade performance and describe (in no more than 10 sentences each) how they work and what their effect is.

Question 5:

Consider a 2D unsteady airfoil. Explain the relation between loads and circulation for the unsteady case and how it differs from the steady case. Consider the wake of the airfoil; explain its origin. Judge the validity of the 2D model in relation to Helmholtz theorem that states that vortices must exist in closed 3D rings.

Question 6:

Discuss the impact of yaw misalignment in a Horizontal Axis Wind Turbine in terms of the flow, loads, wake and power coefficient.

Question 7:

Consider an actuator disk with radius R , operating in a wind field. The wind field is normal to the disk and the velocity magnitude at an infinite distance upwind is U_{inf} . Over the surface of this actuator disk, there is a uniform force field of distributed magnitude f_n , normal to the surface and against the wind.

.1 Describe how velocity, pressure and total pressure vary along the streamtube that intersects the actuator disk, and the expansion or contraction of the streamtube.

.2 Describe the vorticity field in the flow.

.3 Determine expression for the velocity at the actuator, power converted and thrust as a function of f_n .

.4 Consider an additional force distribution over the surface of the actuator, in azimuthal direction, with uniform magnitude f_{azim} . Explain the impact of this additional force distribution in the previous answers in points 1, 2 and 3.